

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

19CSM104 - NATURAL LANGUAGE PROCESSING WITH CLASSIFICATION AND VECTOR SPACES

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) With a suitable example and diagram explain Supervised Machine learning.	10	L*2	1	1
	b) What is Sentiment Analysis? How does Sentiment Analysis work? Explain types of Sentiment Analysis.	10	L1	1	1
2.	a) Explain feature extraction in sentiment analysis with the help of suitable example.	10	L2	1	1
	b) Explain various steps of training Logistic Regression with a suitable diagram.	10	L2	1	1,2
3.	a) Explain the step by step procedure of pre-processing.	10	L2	2	1,2
	b) Explain Bayes Rule and Laplace smoothing in Naive Bayes algorithm.	10	L2	2	1,2
Unit – II					
4.	a) With an example, explain five steps for training a Naive Bayes model.	10	L1	2	1
	b) List and explain Applications of Naive Bayes Algorithm.	10	L2	2	1,2
5.	a) Summarize Euclidean distance. Develop Python code for Euclidean distance.	10	L3	3	1,2
	b) Write and explain PCA Algorithm.	10	L2	3	2
6.	a) What are vector space models? Explain its Applications.	10	L1	3	2
	b) What are important tools available for developing NLP applications? Write the feature of each tool.	10	L1	3	2
Unit – III					
7.	a) Explain K-Nearest Neighbor algorithm. List its advantages and disadvantages.	10	L2	4	1,2
	b) What is hashing table? Explain hashing functions.	10	L2	4	2
8.	a) With suitable example elaborate Locality Sensitive Hashing algorithm.	10	L3	4	2
	b) Explain machine translation using the example of english to french translation.	10	L2	3	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
